Lab Work 6: Data Aggregation and Reporting with SQL Stored Procedures

Objective: Learn how to aggregate, analyze, and report branch and population data by writing SQL queries and automating them within a stored procedure. Understand key SQL concepts like grouping, aggregation, filtering, unions, and ranking.

Tasks:

1. **Total Branches by Province**: Write a query to calculate the total number of branches in each province by summing the "Total" column grouped by province.

```sql
SELECT Province, SUM(Total) AS TotalBranches
FROM MonthlyStatistics
GROUP BY Province;
```

100 % No issues found

Results | Messages

	Province	TotalBranches
1	Bagmati	2286
2	Gandaki	834
3	Karnali	240
4	Koshi	994
5	Lumbini	1048
6	Madhesh	717
7	Sudur Paschim	388

2. **Average Population per Branch by Province**: Write a query to find the average population per branch for each province using the "Pop._Per_Branch" column.

```sql
SELECT Province, AVG([Population_Per_Branch]) AS AvgPopPerBranch
FROM MonthlyStatistics
GROUP BY Province;

select * from MonthlyStatistics
```

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Results | Messages

	Province	AvgPopPerBranch
1	Bagmati	4279.063076
2	Gandaki	3005.059090
3	Karnali	7037.635000
4	Koshi	5121.981428
5	Lumbini	5840.412500
6	Madhesh	8750.910000
7	Sudur Paschim	8153.073333

3. **Identify Districts with Inconsistent Branch Counts**: Find districts where the sum of Class_A, Class_B, and Class_C does not match the value in the "Total" column, indicating data inconsistency.

```sql
SELECT
    Province,
    District,
    Class_A,
    Class_B,
    Class_C,
    (Class_A + Class_B + Class_C) AS ComputedTotal,
    Total AS ReportedTotal
FROM MonthlyStatistics
WHERE (Class_A + Class_B + Class_C) != Total;
```

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Results Messages

Province	District	Class_A	Class_B	Class_C	ComputedTotal	ReportedTotal

4. **Provinces with More than 10 Branches**: Write a query to list provinces where the total number of branches exceeds 100, using SQL's HAVING clause.

```sql
SELECT Province, SUM(Total) AS TotalBranches
FROM MonthlyStatistics
GROUP BY Province
HAVING SUM(Total) > 100;
```

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Results Messages

	Province	TotalBranches
1	Bagmati	2286
2	Gandaki	834
3	Karnali	240
4	Koshi	994
5	Lumbini	1048
6	Madhesh	717
7	Sudur Paschim	388

5. **List Provinces and Districts with Type Labels**: Create a combined list of all provinces and districts, tagging each row with a type label — either "Province" or "District".

```sql
SELECT DISTINCT Province AS Name, 'Province' AS Type
FROM MonthlyStatistics

UNION

SELECT District, 'District'
FROM MonthlyStatistics;
```

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esults Messages

Name	Type
Bagmati	Province
Gandaki	Province
Karnali	Province
Koshi	Province
Lumbini	Province
Madhesh	Province
Sudur Paschim	Province
Achham	District
Arghakhanchi	District
Baglung	District
Baitadi	District

6. **Districts with High Branch Counts in Specific Classes**: Identify districts where Class_A branches exceed 10 or Class_B branches exceed 15, showcasing filtering with OR conditions and UNION.

```sql
SELECT Province, District, Class_A, Class_B
FROM MonthlyStatistics
WHERE Class_A > 10

UNION

SELECT Province, District, Class_A, Class_B
FROM MonthlyStatistics
WHERE Class_B > 15;
```

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Results Messages

Province	District	Class_A	Class_B
Bagmati	Bhaktapur	125	19
Bagmati	Chitawan	167	53
Bagmati	Dhading	62	17
Bagmati	Dolakha	31	3
Bagmati	Kathmandu	936	150
Bagmati	Kavrepalanchok	70	14
Bagmati	Lalitpur	235	34
Bagmati	Makwanpur	56	14
Bagmati	Nuwakot	40	15
Bagmati	Ramechhap	22	2
Bagmati	Rasuwa	28	1

7. **District with Highest Population in Each Province**: Write a query to find the district(s) with the highest population in every province.

```sql
SELECT Province, District, Population
FROM MonthlyStatistics AS a
WHERE Population = (
    SELECT MAX(Population)
    FROM MonthlyStatistics AS b
    WHERE a.Province = b.Province
);
```

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Results | Messages

Province	District	Population
Sudur Paschim	Kailali	904666
Madhesh	Dhanusa	867747
Lumbini	Rupandehi	1121957
Koshi	Morang	1148156
Karnali	Surkhet	415126
Gandaki	Kaski	600051
Bagmati	Kathmandu	2041587

8. **Top 3 Districts with the Highest Number of Branches Overall**: Retrieve the top three districts ranked by total branches across all provinces.

```sql
SELECT TOP 3 Province, District, Total
FROM MonthlyStatistics
ORDER BY Total DESC;
```

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Results | Messages

Province	District	Total
Bagmati	Kathmandu	1135
Lumbini	Rupandehi	325
Gandaki	Kaski	301

9. **Number of Districts and Average Population per Province**: For each province, calculate the number of districts it contains and the average population across those districts.

```sql
SELECT Province, COUNT(DISTINCT District) AS DistrictCount, AVG(Population)
FROM MonthlyStatistics
GROUP BY Province;
```

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Results | Messages

	Province	DistrictCount	AvgPopulation
1	Bagmati	13	470528
2	Gandaki	11	224220
3	Karnali	10	168841
4	Koshi	14	354386
5	Lumbini	12	426839
6	Madhesh	8	764325
7	Sudur Paschim	9	299420

10. **Provinces with Average Branches per District Greater Than 20**: Identify provinces where the average number of branches per district exceeds 20.

```
SELECT Province
FROM MonthlyStatistics
GROUP BY Province
HAVING (SUM(Total) * 1.0 / COUNT(DISTINCT District)) > 20;
```

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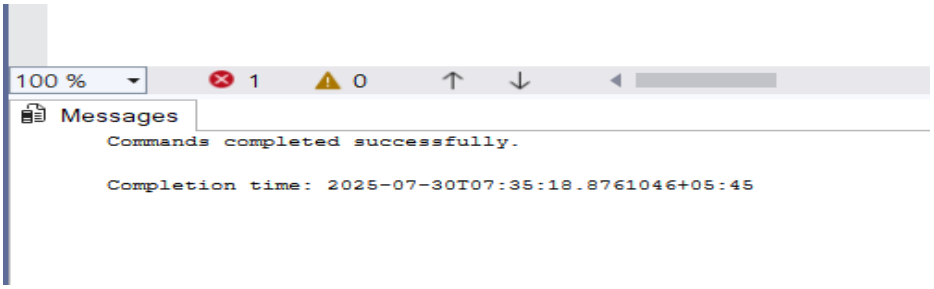
Results | Messages

	Province
1	Bagmati
2	Gandaki
3	Karnali
4	Koshi
5	Lumbini
6	Madhesh
7	Sudur Paschim

11. Combine all above queries into a single stored procedure that inserts their results into corresponding summary tables for automated reporting.

```
CREATE OR ALTER PROCEDURE GenerateBranchPopulationReports
AS
BEGIN
    -- 1. Total Branches by Province
    IF OBJECT_ID('BranchesByProvince') IS NOT NULL DROP TABLE BranchesByProvince;
    SELECT Province, SUM(Total) AS TotalBranches
    INTO BranchesByProvince
    FROM nrb_monthly_statistics
    GROUP BY Province;

    -- 2. Average Population per Branch by Province
    IF OBJECT_ID('AvgPopulationPerBranch') IS NOT NULL DROP TABLE AvgPopulationPerBranch;
    SELECT Province, AVG([Pop_Per_Branch]) AS AvgPopPerBranch
    INTO AvgPopulationPerBranch
    FROM nrb_monthly_statistics
    GROUP BY Province;
```

```
-- 3. Inconsistent Branch Counts
IF OBJECT_ID('InconsistentBranchCounts') IS NOT NULL DROP TABLE InconsistentBranchCounts;
SELECT
    Province,
    District,
    Class_A,
    Class_B,
    Class_C,
    Total,
    (Class_A + Class_B + Class_C) AS ComputedTotal
INTO InconsistentBranchCounts
FROM nrb_monthly_statistics
WHERE (Class_A + Class_B + Class_C) != Total;

-- 4. Provinces with More Than 100 Branches
IF OBJECT_ID('ProvincesOver100Branches') IS NOT NULL DROP TABLE ProvincesOver100Branches;
SELECT Province, SUM(Total) AS TotalBranches
INTO ProvincesOver100Branches
FROM nrb_monthly_statistics
GROUP BY Province
HAVING SUM(Total) > 100;

-- 5. Provinces and Districts with Type Labels
IF OBJECT_ID('ProvinceDistrictList') IS NOT NULL DROP TABLE ProvinceDistrictList;
SELECT DISTINCT Province AS Name, 'Province' AS Type
INTO ProvinceDistrictList
FROM nrb_monthly_statistics
UNION
```

```
SELECT DISTINCT District, 'District'
FROM nrb_monthly_statistics;

-- 6. Districts with High Branch Counts in Specific Classes
IF OBJECT_ID('HighClassBranchDistricts') IS NOT NULL DROP TABLE HighClassBranchDistricts;
SELECT Province, District, Class_A, Class_B
INTO HighClassBranchDistricts
FROM nrb_monthly_statistics
WHERE Class_A > 10
UNION
SELECT Province, District, Class_A, Class_B
FROM nrb_monthly_statistics
WHERE Class_B > 15;

-- 7. District with Highest Population in Each Province
IF OBJECT_ID('TopPopulationDistricts') IS NOT NULL DROP TABLE TopPopulationDistricts;
SELECT Province, District, Population
INTO TopPopulationDistricts
FROM nrb_monthly_statistics AS t
WHERE Population = (
   SELECT MAX(Population)
   FROM nrb_monthly_statistics
   WHERE Province = t.Province
);

-- 8. Top 3 Districts with Highest Total Branches
IF OBJECT_ID('Top3BranchDistricts') IS NOT NULL DROP TABLE Top3BranchDistricts;
SELECT TOP 3 District, Province, SUM(Total) AS TotalBranches
```

```
    INTO Top3BranchDistricts
    FROM nrb_monthly_statistics
    GROUP BY Province, District
    ORDER BY SUM(Total) DESC;

    -- 9. Number of Districts and Average Population per Province
    IF OBJECT_ID('DistrictStatsByProvince') IS NOT NULL DROP TABLE DistrictStatsByProvince;
    SELECT Province, COUNT(DISTINCT District) AS DistrictCount, AVG(Population) AS AvgPopulation
    INTO DistrictStatsByProvince
    FROM nrb_monthly_statistics
    GROUP BY Province;

    -- 10. Provinces with Avg Branches per District > 20
    IF OBJECT_ID('HighAvgBranchProvinces') IS NOT NULL DROP TABLE HighAvgBranchProvinces;
    SELECT Province,
        COUNT(DISTINCT District) AS DistrictCount,
        SUM(Total) AS TotalBranches,
        (1.0 * SUM(Total)) / COUNT(DISTINCT District) AS AvgBranchesPerDistrict
    INTO HighAvgBranchProvinces
    FROM nrb_monthly_statistics
    GROUP BY Province
    HAVING (1.0 * SUM(Total)) / COUNT(DISTINCT District) > 20;
END;
```